Lab 7: Fuzz Testing with Hypothesis

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Assigned repo: SETAPESU26/L6_L7_42_salon_booking (Lab 7 folder)

Tools: Python 3.12.0, Hypothesis 6.170.15, pytest 9.2.0

The assigned script is `salon_booking.py` (the handout calls it `processor.py`) and the test file is `test_salon_booking.py`. It has three functions: `parse_booking`, `duration_average` and `payable_bill`.

Screenshots were drawn from the real output of each command, run in the Lab-7 folder.

Step 1: Install dependencies

```
pip install hypothesis pytest
```

```
PS Lab-7> pip install hypothesis pytest
Requirement already satisfied: hypothesis in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (6.170.15)
Requirement already satisfied: pytest in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (9.2.0)
Requirement already satisfied: sortedcontainers<3.0.0,>=2.1.0 in
c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from hypothesis) (2.4.0)
Requirement already satisfied: colorama>=0.4 in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from
pytest) (0.4.6)
Requirement already satisfied: iniconfig>=2 in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from
pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from
pytest) (25.0)
Requirement already satisfied: pluggy<2,>=1.6 in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from
pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in c:\users\sripr\appdata\local\programs\python\python312\lib\site-packages (from
pytest) (2.21.0)
WARNING: Ignoring invalid distribution ~ip (C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ip (C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ip (C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages)
```

Hypothesis and pytest installed.

Step 2: Run the original script

```
python salon_booking.py
```

With the example inputs the script works. With other inputs it either accepts nonsense (a booking of -30 minutes, an average of -20 minutes, a bill of *nan*) or crashes with a `ValueError`.

```
PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,45,60
Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 800|100
Payable Bill: 844.0

PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): Spa:-30
Service and Duration: ('Spa', -30.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 10,-50
Average Duration: -20

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): nan|0
Payable Bill: nan

PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut
Traceback (most recent call last):
  File "D:\College\sem 5\SE_mini_project\Lab-7\salon_booking.py", line 31, in <module>
    main()
  File "D:\College\sem 5\SE_mini_project\Lab-7\salon_booking.py", line 17, in main
    result = parse_booking(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "D:\College\sem 5\SE_mini_project\Lab-7\salon_booking.py", line 2, in parse_booking
    service, minutes = text.split(':')
    ^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Original script: a normal run, a run that gives wrong results, and a run that crashes.

Step 3: Complete test_salon_booking.py with Hypothesis

The starter file only had the three empty test headers. Each function now has four tests, 12 in all:

• **no_crash**: any text, `None`, or a string built from tricky pieces (`nan`, `inf`, `1e400`, `-5`, separators, an Arabic digit) must not raise an exception.
• **output_is_valid**: for the same inputs, the result must be `None` (rejected) or a sensible value: a finite number, inside the allowed range, with a non-empty service name.
• **valid_input**: generated well-formed input must give the correct answer, so a fix that rejects everything cannot pass.
• **rejects_***: zero or negative minutes and negative discounts must be rejected.

Each test runs 300 generated examples. The code is in `test_salon_booking.py`.

Step 4: Run the tests against the original script

```
python -m pytest test_salon_booking.py
```

10 of the 12 tests fail. Only the two **valid_input** tests for parse_booking and duration_average pass.

```
PS Lab-7> python -m pytest test_salon_booking.py --tb=no -rf
============================= test session starts =============================
platform win32 -- Python 3.12.0, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\College\sem 5\SE_mini_project\Lab-7
plugins: anyio-4.14.2, hypothesis-6.170.15
collected 12 items

test_salon_booking.py FF.FFF.FFFFF                                       [100%]

============================== warnings summary ===============================
C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning:
_pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ===========================
FAILED test_salon_booking.py::test_parse_booking_no_crash - ExceptionGroup: H...
FAILED test_salon_booking.py::test_parse_booking_output_is_valid - ExceptionG...
FAILED test_salon_booking.py::test_parse_booking_rejects_out_of_range_minutes
FAILED test_salon_booking.py::test_duration_average_no_crash - ExceptionGroup...
FAILED test_salon_booking.py::test_duration_average_output_is_valid - Excepti...
FAILED test_salon_booking.py::test_duration_average_rejects_non_positive - As...
FAILED test_salon_booking.py::test_payable_bill_no_crash - ExceptionGroup: Hy...
FAILED test_salon_booking.py::test_payable_bill_output_is_valid - ExceptionGr...
FAILED test_salon_booking.py::test_payable_bill_valid_input - assert 0.179999...
FAILED test_salon_booking.py::test_payable_bill_rejects_negative_discount - A...
================== 10 failed, 2 passed, 1 warning in 5.58s ==================
```

Summary of the failing run.

```
PS Lab-7> python -m pytest test_salon_booking.py   # excerpt: crashes found
_____________________ test_payable_bill_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "test_salon_booking.py", line 111, in test_payable_bill_no_crash
  |     def test_payable_bill_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 113, in test_payable_bill_no_crash
    |     payable_bill(s)
    |   File "salon_booking.py", line 11, in payable_bill
    |     total = float(bill) * 1.18 - float(discount)
    |             ^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 115, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e!r}"
    | AssertionError: payable_bill crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 113, in test_payable_bill_no_crash
    |     payable_bill(s)
    |   File "salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 115, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e!r}"
    | AssertionError: payable_bill crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 113, in test_payable_bill_no_crash
    |     payable_bill(s)
    |   File "salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_salon_booking.py", line 115, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e!r}"
    | AssertionError: payable_bill crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

Excerpt: Hypothesis found three different crashes in payable_bill and shrank each to the smallest input.

```
PS Lab-7> python -m pytest test_salon_booking.py   # excerpt: wrong results found
_____________ test_parse_booking_rejects_out_of_range_minutes _____________

    @given(service_names, st.integers(max_value=0) | st.integers(min_value=MAX_MINUTES + 1))
>   def test_parse_booking_rejects_out_of_range_minutes(service, minutes):
                ^^^

test_salon_booking.py:71:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

service = 'A', minutes = 0

    @given(service_names, st.integers(max_value=0) | st.integers(min_value=MAX_MINUTES + 1))
    def test_parse_booking_rejects_out_of_range_minutes(service, minutes):
>       assert parse_booking(f"{service}:{minutes}") is None
E       AssertionError: assert ('A', 0.0) is None
E        +  where ('A', 0.0) = parse_booking('A:0')
E       Failing test case: test_parse_booking_rejects_out_of_range_minutes(
E           service='A',
E           minutes=0,
E       )

test_salon_booking.py:72: AssertionError
_____________ test_duration_average_rejects_non_positive _____________

    @given(st.lists(st.integers(1, MAX_MINUTES), max_size=5), st.integers(max_value=0))
>   def test_duration_average_rejects_non_positive(values, bad):
                ^^^

test_salon_booking.py:103:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

values = [], bad = 0

    @given(st.lists(st.integers(1, MAX_MINUTES), max_size=5), st.integers(max_value=0))
    def test_duration_average_rejects_non_positive(values, bad):
        csv = ",".join(str(v) for v in values + [bad])
>       assert duration_average(csv) is None
E       AssertionError: assert 0 is None
E        +  where 0 = duration_average('0')
E       Failing test case: test_duration_average_rejects_non_positive(
E           values=[],
E           bad=0,
E       )

test_salon_booking.py:105: AssertionError
```

Excerpt: tests that found wrong results rather than crashes (parse_booking, duration_average).

```
PS Lab-7> python -m pytest test_salon_booking.py   # excerpt: wrong results found
_____________________ test_payable_bill_valid_input _____________________

    @given(st.integers(0, 1_000_000), st.integers(0, 1_000_000))
>   def test_payable_bill_valid_input(bill, discount):
                ^^^

test_salon_booking.py:128:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

bill = 1, discount = 1

    @given(st.integers(0, 1_000_000), st.integers(0, 1_000_000))
    def test_payable_bill_valid_input(bill, discount):
        result = payable_bill(f"{bill}|{discount}")
>       assert result == round(max(bill * 1.18 - discount, 0), 2)
E       assert 0.17999999999999994 == 0.18
E        +  where 0.18 = round(0.17999999999999994, 2)
E        +    where 0.17999999999999994 = max(((1 * 1.18) - 1), 0)
E       Failing test case: test_payable_bill_valid_input(
E           bill=1,
E           discount=1,
E       )

test_salon_booking.py:130: AssertionError
_____________ test_payable_bill_rejects_negative_discount _____________

    @given(st.integers(0, 1_000_000), st.integers(max_value=-1))
>   def test_payable_bill_rejects_negative_discount(bill, discount):
                ^^^

test_salon_booking.py:135:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

bill = 0, discount = -1

    @given(st.integers(0, 1_000_000), st.integers(max_value=-1))
    def test_payable_bill_rejects_negative_discount(bill, discount):
>       assert payable_bill(f"{bill}|{discount}") is None
E       AssertionError: assert 1.0 is None
E        +  where 1.0 = payable_bill('0|-1')
E       Failing test case: test_payable_bill_rejects_negative_discount(
E           bill=0,
E           discount=-1,
E       )

test_salon_booking.py:136: AssertionError
```

Excerpt: tests that found wrong results rather than crashes (payable_bill).

Step 5: Fix the script

Every input that Hypothesis reported, and what was changed:

Function	Input found	Original behaviour	Fix
All three	None	AttributeError on .split	Return None if input is not a string
parse_booking	'' (no colon)	ValueError: not enough values to unpack	Require exactly one ':'
parse_booking	':'	ValueError: float('')	Convert inside try/except
parse_booking	'0:1E400'	Returns inf minutes	Reject nan and inf
parse_booking	'0:0', 'A:0'	Accepts 0 minutes	Minutes must be above 0 and up to 1440
parse_booking	':0'	Accepts an empty service name	Name must contain a letter
duration_average	''	ValueError: int('')	Convert inside try/except
duration_average	'0'	Average of 0 minutes	Each value must be 1 to 1440
payable_bill	''	ValueError: not enough values to unpack	Require exactly one '|'
payable_bill	'|'	ValueError: float('')	Convert inside try/except
payable_bill	'0|-1E400'	Returns inf	Reject nan and inf
payable_bill	'0|-1'	Negative discount raises the bill to 1.0	Amounts must be 0 to 10,000,000
payable_bill	'1|1'	0.17999999999999994	Round to 2 decimal places

Invalid input now gives `None` instead of an exception, and `main()` prints a short message saying what format is expected. The 1440-minute limit (one day) and the 10,000,000 limit on amounts also stop a huge number from overflowing to *inf* after GST is added.

```
python salon_booking.py
```

```
PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,45,60
Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 800|100
Payable Bill: 844.0

PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): Spa:-30
Invalid booking. Use service:minutes, with minutes above 0 and up to 1440.

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 10,-50
Invalid durations. Use whole minutes from 1 to 1440, separated by commas.

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): nan|0
Invalid bill. Use bill|discount with amounts from 0 to 10000000.

PS Lab-7> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut
Invalid booking. Use service:minutes, with minutes above 0 and up to 1440.

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60):
Invalid durations. Use whole minutes from 1 to 1440, separated by commas.

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): |
Invalid bill. Use bill|discount with amounts from 0 to 10000000.
```

Fixed script: the normal run gives the same results; bad input is rejected with a message instead of a crash.

Step 6: Re-run the tests against the fixed script

```
python -m pytest test_salon_booking.py
```

```
PS Lab-7> python -m pytest test_salon_booking.py
============================= test session starts =============================
platform win32 -- Python 3.12.0, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\College\sem 5\SE_mini_project\Lab-7
plugins: anyio-4.14.2, hypothesis-6.170.15
collected 12 items

test_salon_booking.py ............                                       [100%]

============================== warnings summary ===============================
C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning:
_pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 12 passed, 1 warning in 3.45s ========================
```

All 12 tests pass on the fixed script. The warning comes from the installed anyio plugin, not from this lab's code.

The tests were also run with five different random seeds (`--hypothesis-seed=1` to 5); all 12 passed each time.

Deliverables

1. This PDF with the screenshots
2. Fixed script: `Lab-7/salon_booking.py`
3. Updated tests: `Lab-7/test_salon_booking.py`